CALCULUS (II)

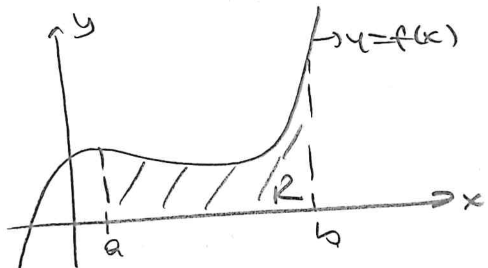
WEEK (XII)

MULTIPLE INTEGRALS

DOUBLE INTEGRALS

— The area problem motivates the definition of the definite integral

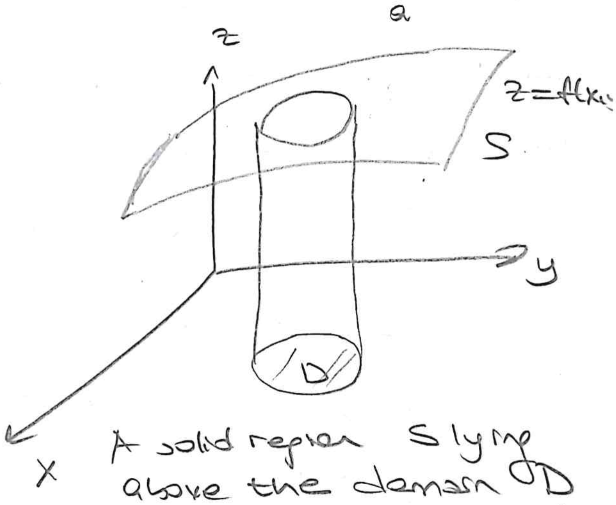
$$A = \int_a^b f(x)\,dx$$



The area under the function $f(x)=y$ for $x \in [a,b] = \int_a^b f(x)\,dx$

— The volume problem motivates the definition of the double integral

$$V = \iint_D f(x,y)\,dA$$

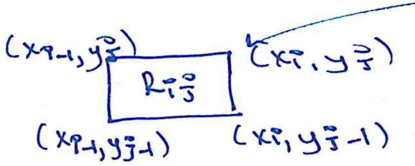


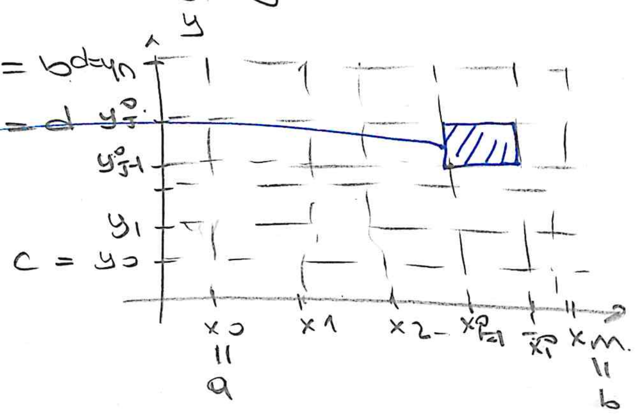
A solid region S lying above the domain D in the xy-plane and below the surface z=f(x,y)

→ Let us start with the case where D is a closed rectangle with sides parallel to the coordinate axes in the xy-plane, and f is a bounded function on D. If D consists of the points (x,y) such that $a \le x \le b$ and $c \le y \le d$, we can form a partition P of D into small rectangles by partitioning each of intervals $[a,b]$ and $[c,d]$, say by points

$$a = x_0 < x_1 < x_2 < \cdots < x_{m-1} < x_m = b$$

$$c = y_0 < y_1 < y_2 < \cdots < y_{n-1} < y_n = d$$





→ The rectangle R_{ij} has area

$$\Delta A_{ij} = \Delta x_i \Delta y_j = (x_i - x_{i-1})(y_j - y_{j-1})$$

and diameter (diagonal length)

$$\operatorname{diam}(R_{ij}) = \sqrt{(\Delta x_i)^2 + (\Delta y_j)^2} = \sqrt{(x_i - x_{i-1})^2 + (y_j - y_{j-1})^2}$$

The <u>norm</u> of the partition P is the largest of these subrectangle diameters.

$$\|P\| = \max_{\substack{1 \le i \le m \\ 1 \le j \le n}} \operatorname{diag}(R_{ij})$$

→ Now, we pick an arbitrary point (x_{ij}^{*}, y_{ij}^{*}) in each of rectangles R_{ij} and form the Riemann sum

$$\boxed{R(f,P) = \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^{*}, y_{ij}^{*})\, \Delta A_{ij}}$$

which is the sum of $m \cdot n$ terms, one for each rectangle in the partition.

→ The double integral of f over D is defined to be the limit of such Riemann sums, provided the limit exists as $\|P\| \to 0$ independently of how the points (x_{ij}^{*}, y_{ij}^{*}) are chosen.

Definition The double integral over a rectangle.

We say that f is integrable over the rectangle D and has double integral

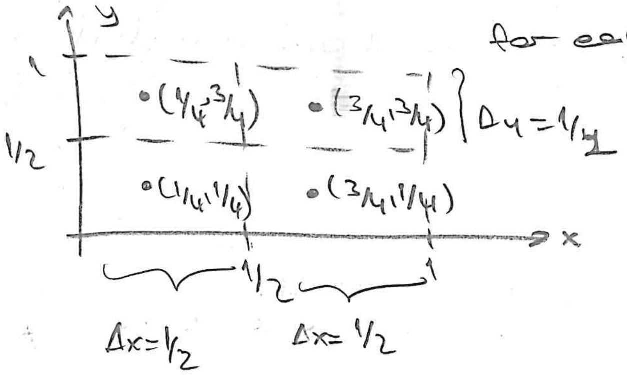
$$\boxed{I = \iint_D f(x,y)\, dA}$$

If for every positive number ε there exists a number δ depending on ε, such that

$$\boxed{|R(f,P) - I| < \varepsilon}$$

holds for every partition P of D satisfying $\|P\| < \delta$ and for all choices of the points (x_{ij}^*, y_{ij}^*) in the subrectangles of P.

Ex 7: Let D be the square $0 \le x \le 1$, $0 \le y \le 1$. Use Rieman sum corresponding to the partition of D into four squares with points selected at the center of rectangles. compute $\iint_D (x^2+y)\, dA$.



for each rectangle $\Delta x = 1/2$ and $\Delta y = 1/2$.

The area of each rectangle.

$$\Delta A = \Delta x \cdot \Delta y = 1/4$$

The center of squares are $(1/4, 1/4)$, $(1/4, 3/4)$, $(3/4, 1/4)$ and $(3/4, 3/4)$

$$\iint_D (x^2+y)\, dA \approx R(x^2+y, P) = \underbrace{\left(\tfrac{1}{16} + \tfrac{1}{4}\right)}_{\substack{\text{first rectangle} \\ (x,y) = (1/4,\, 1/4)}} \cdot \tfrac{1}{4} + \underbrace{\left(\tfrac{1}{16} + \tfrac{3}{4}\right)}_{(x,y) = (1/4,\, 3/4)} \cdot \tfrac{1}{4}$$

$$+ \underbrace{\left(\tfrac{9}{16} + \tfrac{1}{4}\right)}_{(x,y) = (3/4,\, 1/4)} \tfrac{1}{4} + \underbrace{\left(\tfrac{9}{16} + \tfrac{3}{4}\right)}_{(x,y) = (3/4,\, 3/4)} \cdot \tfrac{1}{4} = \frac{13}{16}$$

Double Integrals over More General Domains

+ If $f(x,y)$ is defined and bounded on domain D, let $\hat{f}$ be the extension of f that is zero everywhere outside D:

$$\hat{f}(x,y) = \begin{cases} f(x,y), & \text{if } (x,y) \text{ belongs to } D \\ 0, & \text{if } (x,y) \text{ doesn't belong to } D \end{cases}$$

If D is a bounded domain, then it is contained in some rectangle R with sides parallel to the coordinate axes. If $\hat{f}$ is integrable over R, we say that f is integrable over D and define the double integral of f over D to be

$$\boxed{\iint_D f(x,y)\,dA = \iint_R \hat{f}(x,y)\,dA}$$

THM If f is continuous on a closed, bounded domain D whose boundary consists of finitely many curves of finite length then f is integrable on D.

PROPERTIES OF THE DOUBLE INTEGRAL

If f and g are integrable over D, and if L and M are constants then

1. $\iint_D f(x,y)\,dA = 0$ if D has zero area
2. Area of domain: $\iint_D 1\,dA$ = area of D (because it is the volume of a cylinder with base D and height 1)
3. Integrals representing volumes: If $f(x,y) \geq 0$ on D, then $\iint_D f(x,y)\,dA = V \geq 0$ where V is the volume of the solid lying vertically above D and below the surface $z=f(x,y)$

④ If $f(x,y) \le 0$ on D, then $\iint_D f(x,y)\,dA = -V \le 0$, where V is the volume of the solid lying vertically below D and above the surface $z=f(x,y)$

⑤ Linear dependence on the integrand

$$\iint_D \left(L f(x,y) + M g(x,y)\right) dA = L \iint_D f(x,y)\,dA + M \iint_D g(x,y)\,dA$$

⑥ Inequalities are preserved

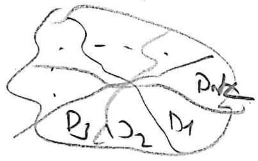
If $f(x,y) \le g(x,y)$ on D, then $\iint_D f(x,y)\,dA \le \iint_D g(x,y)\,dA$.

⑦ The triangle inequality $\left| \iint_D f(x,y)\,dA \right| \le \iint_D |f(x,y)|\,dA$

⑧ Additivity of domains If $D_1, D_2, \ldots, D_k$ are nonoverlapping domains on each of which f is integrable, then f is integrable over the union $D = D_1 \cup D_2 \cup \cdots \cup D_k$, and

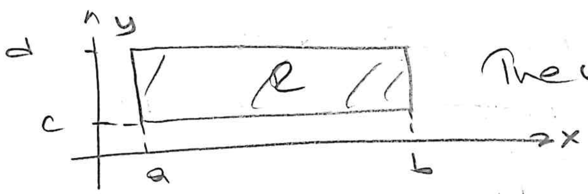
$$\iint_D f(x,y)\,dA = \sum_{j=1}^{k} \iint_{D_j} f(x,y)\,dA.$$

Nonoverlapping domains can share boundaries but have no interior points in common.



Ex If R is the rectangle $a \le x \le b$, $c \le y \le d$, then

$$\iint_R 3\,dA = 3 \times \text{area of } R = 3(b-a)(d-c)$$



The base of a rectangular box.

Ex
7 Evaluate $I = \iint\limits_{x^2+y^2\le 1} (\sin x + y^3 + 4)\,dA$

We can write

$$I = \iint\limits_{x^2+y^2\le 1} \sin x\, dA + \iint\limits_{x^2+y^2\le 1} y^3\, dA + \iint\limits_{x^2+y^2\le 1} 4\,dA.$$

$$= I_1 + I_2 + I_3$$

→ The domain D: $x^2+y^2\le 1$

→ Function $f(x,y) = \sin x$ is an odd function of x, its graph bounds as much volume below xy-plane in the region $x<0$ as it does above the xy-plane in the region $x>0$.

$$\iint\limits_{x^2+y^2\le 1} \sin x\, dA = \underbrace{\iint\limits_{\substack{x^2+y^2\le 1\\ x<0}} \sin x\, dA}_{-V} + \underbrace{\iint\limits_{\substack{x^2+y^2\le 1\\ x>0}} \sin x\, dx}_{V} = 0. = I_1$$

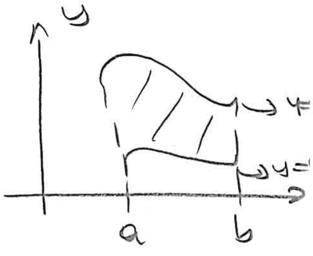
→ $I_2 = \iint\limits_{x^2+y^2\le 1} y^3\, dA = 0$ $f(x,y) = y^3$ is also odd function and D is symetric about the x-axis so $I_2 = 0$.

→ $I_3 = \iint\limits_{D} 4\, dA = 4 \times \text{area of } D = 4\pi$ Thus $I = 0 + 0 + 4\pi = 4\pi$

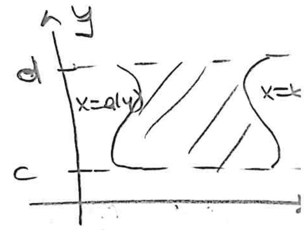
ITERATION OF DOUBLE INTEGRALS IN CARTESIAN COORDINATES

y-simple domain

We say that the domain D in the xy plane is y-simple if it is bounded by two vertical lines $x=a$ and $x=b$, and two continuous graphs $y=c(x)$ and $y=d(x)$ between these lines

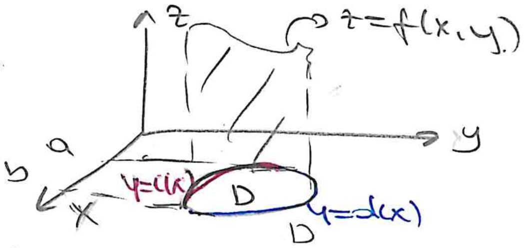


x-simple domain the domain D is bounded by horizontal lines $y=c$ and $y=d$, and two continuous graphs $x=a(y)$ and $x=b(y)$ between these lines



→ Suppose for instance, that D is y-simple and is bounded by $x=a$, $x=b$ and $y=c(x)$ and $y=d(x)$

Then $\iint_D f(x,y)\,dA$ represents (at least for positive f) volume of the solid region inside the vertical cylinder through the boundary of D and between the xy plane and the surface $z=f(x,y)$



The area of the cross-section (cross-section of the solid in the vertical plane perpendicular to x-axis)

$$A(x)=\int_{c(x)}^{d(x)} f(x,y)\,dy$$

$$\boxed{\iint_D f(x,y)\,dA=\int_a^b A(x)\,dx=\int_a^b\left(\int_{c(x)}^{d(x)} f(x,y)\,dy\right)dx}$$

Notationally, it is common to omit the large parentheses and write

$$\iint_D f(x,y)\,dA = \int_a^b \int_{c(x)}^{d(x)} f(x,y)\,dy\,dx = \underbrace{\int_a^b dx \underbrace{\int_{c(x)}^{d(x)} f(x,y)\,dy}_{\text{inner integral}}}_{\text{Iterated integrals.}}$$

→ For double integrals over x-simple domains we can slice perpendicular to the y-axis and obtain an iterated integral with the outer integral in the y-direction.

THM Iteration of Double Integrals

If $f(x,y)$ is continuous on the bounded y-simple domain D given by $a \le x \le b$ and $c(x) \le y \le d(x)$ then

$$\iint_D f(x,y)\,dA = \int_a^b dx \int_{c(x)}^{d(x)} f(x,y)\,dy$$

Similarly, if f is continuous on the x-simple domain D given by $c \le y \le d$ and $a(y) \le x \le b(y)$, then

$$\iint_D f(x,y)\,dA = \int_c^d dy \int_{a(y)}^{b(y)} f(x,y)\,dx$$

NOTATION $\iint_D f(x,y)\,dy\,dx = \iint_D f(x,y)\,dx\,dy = \iint_D f(x,y)\,dA.$

FUBINI'S THEOREM

If $f(x,y)$ is continuous on $R=[a,b]\times[c,d]$ then

$$\boxed{\iint_R f(x,y)\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy}$$

(*) More generally, this is true if we assume that f is bounded on R, f is discontinuous only on a finite number of smooth curves and the iterated integrals exist.

Ex Compute following integrals over the indicated rectangles

(1) $I=\iint_R \left[x^2y^2+\cos(\pi x)+\sin(\pi y)\right]dA \qquad R=\underbrace{[-2,-1]}_{x}\times\underbrace{[0,1]}_{y}$

$$I = \int_0^1 \int_{-2}^{-1} \left[x^2y^2+\cos(\pi x)+\sin(\pi y)\right]dx\,dy$$

$$= \int_0^1 \left[\frac{x^3}{3}y^2 + \frac{\sin(\pi x)}{\pi} + \sin(\pi y)\,x\right]\Big|_{-2}^{-1} dy$$

$$= \int_0^1 \left[\left(-\frac{y^2}{3} + \frac{\cancel{\sin(-\pi)}^{\,0}}{\pi} - \sin(\pi y)\right) - \left(-\frac{8}{3}y^2 + \frac{\cancel{\sin(-2\pi)}^{\,0}}{\pi} - 2\sin(\pi y)\right)\right]$$

$$= \int_0^1 \left(\frac{7}{3}y^2 + \sin(\pi y)\right)dy = \frac{7}{9}y^3 - \frac{\cos(\pi y)}{\pi}\Big|_0^1 = \frac{7}{9} + \frac{1}{\pi} + \frac{1}{\pi}$$

$$= 7/9 + 2/\pi.$$

(2) $\iint_R x e^{xy}\,dA \qquad R=[-1,2]\times[0,1]$

Easier to evaluate w.r.t. y first

$$\iint_R x e^{xy}\,dA = \int_{-1}^{2}\int_0^1 x e^{xy}\,dy\,dx = \int_{-1}^{2} e^{xy}\Big|_0^1 dx$$

$$= \int_{-1}^{2}(e^x - 1)\,dx = e^x - x\Big|_{-1}^{2} = (e^2-2) - (e^{-1}+1)$$

$$= e^2 - e^{-1} - 3.$$

(3) Find the volume of the solid S that is bounded by the elliptic paraboloid $x^2+2y^2+z=16$, the planes $x=2$ and $y=2$ and three coordinate planes $(x=0, y=0, z=0)$

S: the solid lies under $z=16-x^2-2y^2$ and above the square $R=[0,2]\times[0,2]$

$$V=\iint_R (16-x^2-2y^2)\,dA=\int_0^2\int_0^2(16-x^2-2y^2)\,dx\,dy=\int_0^2\left(16x-\frac{x^3}{3}-2xy^2\right)\Big|_0^2 dy$$

$$=\int_0^2\left(32-\frac{8}{3}-4y^2\right)dy=\left(\frac{88}{3}y-\frac{4y^3}{3}\right)\Big|_0^2=\frac{176}{3}-\frac{32}{3}=48$$

(4) Evaluate $\iint_R x\cos^2 y\,dA$ $\quad R=[-2,3]\times[0,\pi/2]$

$$\iint_R x\cos^2(y)\,dA=\left(\int_{-2}^{3}x\,dx\right)\left(\int_0^{\pi/2}\cos^2(y)\,dy\right) \qquad \cos(2y)=2\cos^2 y-1$$

$$=\frac{x^2}{2}\Big|_{-2}^{3}\int_0^{\pi/2}\frac{\cos(2y)+1}{2}\,dy$$

$$=\frac{5}{2}\left(\frac{\sin(2y)}{2}+\frac{y}{2}\right)\Big|_0^{\pi/2}=\frac{5\pi}{8}$$

(5) Find the volume of the solid lying above the square R defined by $0\le x\le 1$ and $1\le y\le 2$ and below the plane $z=4-x-y$

$$V=\int_1^2\int_0^1(4-x-y)\,dx\,dy \quad \text{or} \quad V=\int_0^1\int_1^2(4-x-y)\,dy\,dx$$

<u>I WAY</u> $V=\int_1^2\int_0^1(4-x-y)\,dx\,dy=\int_1^2 dy\int_0^1(4-x-y)\,dx$

$$=\int_1^2\left(4x-\frac{x^2}{2}-yx\right)\Big|_0^1 dy$$

$$=\int_1^2\left(4-\frac{1}{2}-y\right)dy=\frac{7}{2}y-\frac{y^2}{2}\Big|_1^2$$

$$=(7-2)-\left(\tfrac{7}{2}-\tfrac{1}{2}\right)$$

II. WAY

$$V = \int_0^1 \int_1^2 (4-x-y)\,dy\,dx = \int_0^1 \left(4y - xy - \frac{y^2}{2}\right)\Big|_1^2 dx$$

$$= \int_0^1 \left[(8-2y-2) - \left(4 - x - \frac{1}{2}\right)\right] dx$$

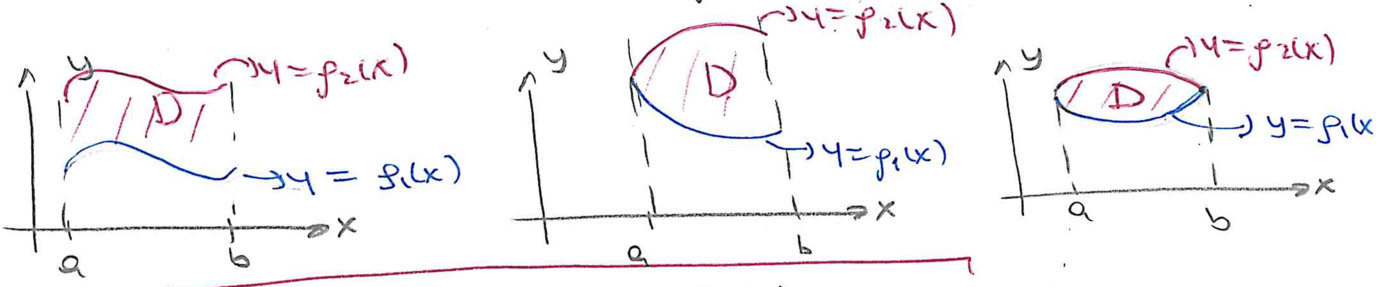
$$= \int_0^1 \left[(6-2x) - \left(7/2 - x\right)\right] dx = \int_0^1 \left(5/2 - x\right) dx = \frac{5}{2}x - \frac{x^2}{2}\Big|_0^1 = 2.$$

DOUBLE INTEGRALS OVER GENERAL REGIONS

Most of the regions are not rectangular

Y-Simple regions: Regions that lies between graphs of two continuous functions of x, that is

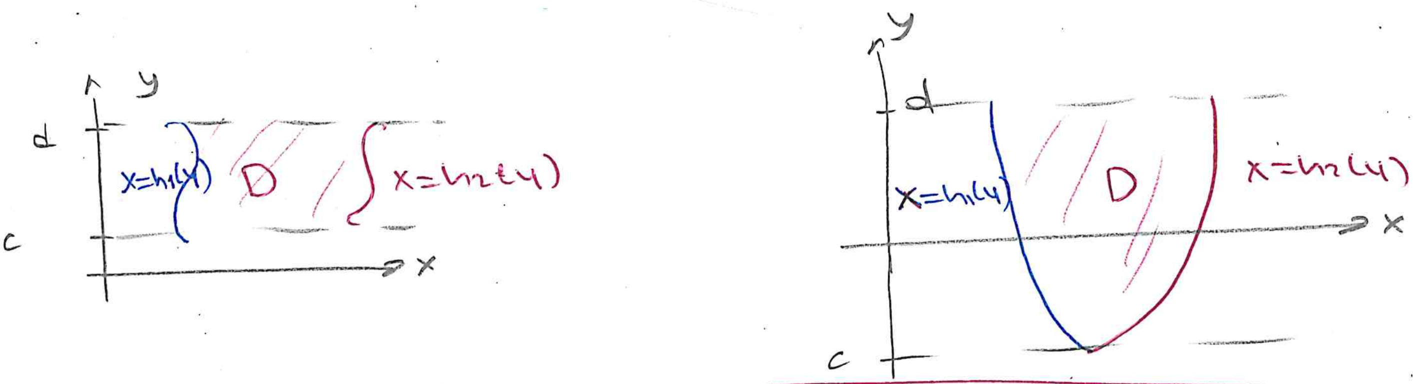
$$D = \{ (x,y) \mid a \le x \le b \quad g_1(x) \le y \le g_2(x) \}$$



$$\boxed{\iint_D f(x,y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\, dy\, dx}$$

X-Simple regions: Region that lies between the graphs of two continuous functions of y, that is

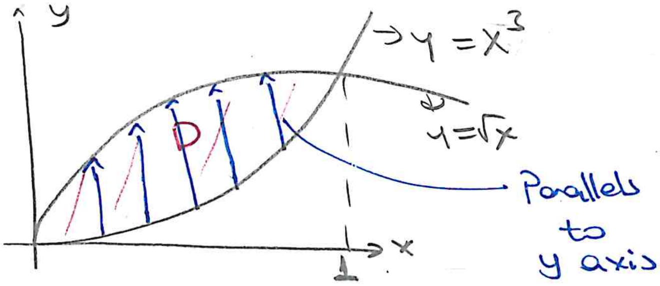
$$D = \{ (x,y) \mid c \le y \le d \quad h_1(y) \le x \le h_2(y) \}$$



$$\boxed{\iint_D f(x,y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\, dx\, dy}$$

Examples Evaluate the following integrals over the given region D

(1) $\iint_D (4xy - y^3)\,dA$ D is the region bounded by $y=\sqrt{x}$ and $y = x^3$



$y = x^3 \wedge y = \sqrt{x}$

$x^3 = \sqrt{x} \qquad x^6 = x$

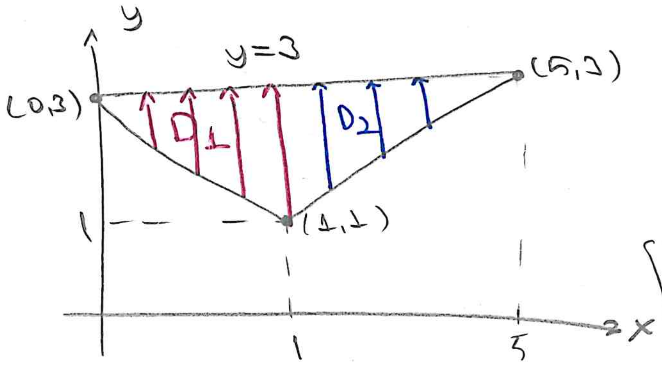
$x(x^5 - 1) = 0 \qquad x = 0,\ x = 1$

$$\iint_D (4xy - y^3)\,dA = \int_0^1 \int_{x^3}^{\sqrt{x}} (4xy - y^3)\,dy\,dx = \int_0^1 \left(2xy^2 - \frac{y^4}{4}\right)\Big|_{x^3}^{\sqrt{x}} dx$$

$$= \int_0^1 \left(2x^2 - \frac{x^2}{4} - 2x^7 + \frac{x^{12}}{4}\right) dx = \left(\frac{7x^3}{12} - \frac{x^8}{4} + \frac{x^{13}}{52}\right)\Big|_0^1 = \frac{7}{12} - \frac{1}{4} + \frac{1}{52} = \boxed{\frac{55}{156}}$$

(2) $\iint_D (6x^2 - 40y)\,dA$ D is the triangle with vertices (0,3), (1,1) and (5,3)

I. WAY



$D = D_1 \cup D_2$

$D_1 = \{(x,y) \mid 0 \le x \le 1 \quad 3 - 2x \le y \le 3\}$

$D_2 = \{(x,y) \mid 1 \le x \le 5 \quad \tfrac{1}{2}x + \tfrac{1}{2} \le y \le 3\}$

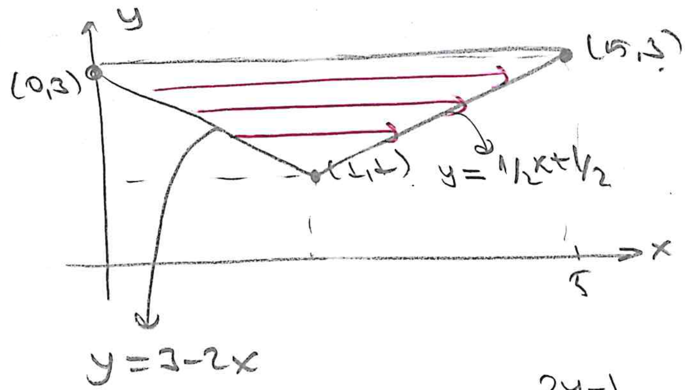
Equation of a line whose two points is given (1,1) (5,3)

$$y - 1 = \frac{3-1}{5-1}(x-1)$$

$$y - 1 = \frac{1}{2}(x-1) \qquad y = \frac{x}{2} + \frac{1}{2}$$

$$\iint_D (6x^2 - 40y)\,dA = \underbrace{\int_0^1 \int_{3-2x}^{3} (6x^2 - 40y)\,dy\,dx}_{\text{Integral on } D_1} + \underbrace{\int_1^5 \int_{\frac{x+1}{2}}^{3} (6x^2 - 40y)\,dy}_{\text{Integral on } D_2}$$

II. WAY



$y = 3-2x \Rightarrow x = \frac{3-y}{2}$

$y = 1/2\,x + 1/2 \Rightarrow x = 2y-1$

$D = \{(x,y): \frac{3-y}{2} \le x \le 2y-1 \quad 1 \le y \le 3\}$

$$\iint_D (6x^2-40y)\,dA = \int_1^3 \int_{\frac{3-y}{2}}^{2y-1} (6x^2-40y)\,dx\,dy = \int_1^3 \left[2x^3-40xy\right]\Big|_{\frac{3-y}{2}}^{2y-1} = -9$$

(3) $\displaystyle\int_0^3 \int_{x^2}^9 x^3 e^{y^3}\,dy\,dx$?

(*) If we try to integrate firstly w.r.t y, that is $dydx$, we can't evaluate it $\int e^{y^3}dy = ?$ so we can't evaluate with order $dydx$ $\displaystyle\int_0^3\int_{x^2}^9 x^3 e^{y^3}\,dy\,dx$

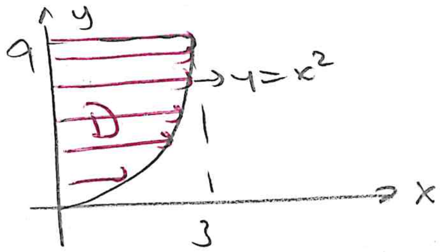
So we must change the order of integral

$y = x^2 \quad y = 9$

$x = 0 \quad x = 3$

$0 \le x \le \sqrt{y}$

$0 \le y \le 9$

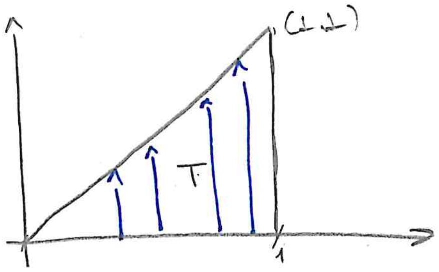


$$\int_0^3\int_{x^2}^9 x^3 e^{y^3}\,dy\,dx = \int_0^9\int_0^{\sqrt{y}} x^3 e^{y^3}\,dx\,dy = \int_0^9 \left\{\frac{x^4}{4} e^{y^3}\Big|_0^{\sqrt{y}}\right\}dy$$

$$= \frac{1}{4}\int_0^9 y^2 e^{y^3}\,dy = \frac{1}{12} e^{y^3}\Big|_0^9 = \frac{1}{12}\left(e^{729}-1\right)$$

Exercise Evaluate $\displaystyle\int_0^8 \int_{\sqrt[3]{y}}^2 \sqrt{x^4+1}\,dx\,dy$

(4) Evaluate $\iint_T xy\,dA$ over the triangle T with vertices $(0,0)$, $(1,0)$ and $(1,1)$



$$\iint_T xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx$$

$$= \int_0^1 \left(\frac{xy^2}{2}\right)\Big|_{y=0}^{y=x} dx$$

$$= \int_0^1 \frac{x^3}{2}\,dx = \frac{x^4}{8}\Big|_0^1 = \frac{1}{8}$$

Compute $\iint_T dA = \int_0^1 \int_0^x dy\,dx = \frac{1}{2} =$ area of triangle T

NOTE If we integrate the constant function $f(x,y)=1$ over a region D, we get the area of D.

$$\boxed{\iint_D 1\,dA = A(D) = \text{area of } D}$$

(6) If $m \le f(x,y) \le M$ for all (x,y) in domain D, then

$$m\,A(D) \le \iint_D f(x,y)\,dA \le M \cdot \underbrace{A(D)}_{\text{Area of } D}$$

(5) Estimate the integral $\iint_D e^{\sin x \cos y}\,dA$ where D is the disk with center origin and radius 2.

Since $-1 \le \sin x \le 1$, $-1 \le \cos y \le 1$ we have

$-1 \le \sin x \cos y \le 1$

Hence, $\underbrace{e^{-1}}_{m} \le e^{\sin x \cos y} \le \underbrace{e}_{M}$

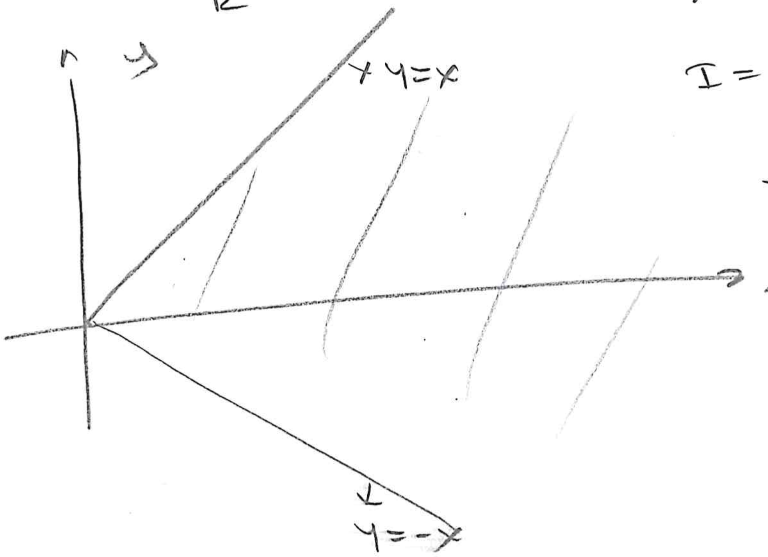
So, we have $e^{-1} A(D) \le \iint e^{\sin x \cos y}\,dA \le e \cdot A(D)$

$A(D) = 4\pi = \pi r^2$. $\quad \frac{4\pi}{e} \le \iint e^{\sin x \cos y}\,dA \le 4\pi e.$

IMPROPER INTEGRALS OF POSITIVE FUNCTIONS

An improper integral of a function f satisfying $f(x,y) \geq 0$ on the domain D must either exist (i.e. converge to a finite value) or be infinite (diverge to infinity)

Ex $I = \iint_R e^{-x^2}\, dA$ where R is the region with $x \geq 0$, $-x \leq y \leq x$



$$I = \int_0^\infty \int_{-x}^{x} e^{-x^2}\, dy\, dx$$

$$= 2\int_0^\infty x e^{-x^2}\, dx$$

$$= 2 \lim_{r \to \infty} \int_0^r x e^{-x^2}\, dx$$

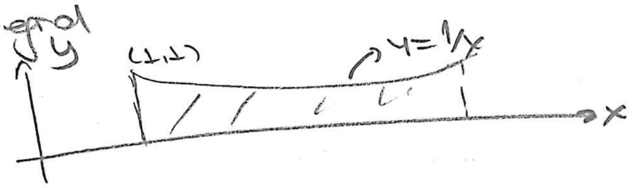
$$= \lim_{r \to \infty} \left(-e^{-x^2}\right)\Big|_0^r$$

$$= \lim_{r \to \infty} \left(1 - e^{-r^2}\right) = 1$$

The given improper integral converges to 1.

Ex If D is the region lying above the x-axis under the curve $y = 1/x$ and to the right of the line $x=1$, determine whether the double integral

$$\iint_D \frac{dA}{x+y}$$



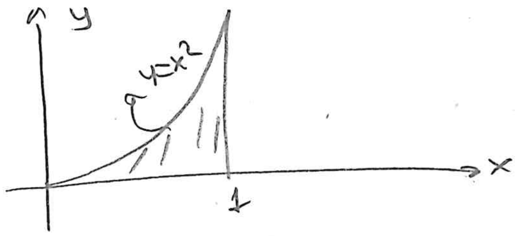
$$\iint_D \frac{dA}{x+y} = \int_1^\infty \int_0^{1/x} \frac{dy}{x+y}\, dx = \int_1^\infty \ln(x+y)\Big|_{y=0}^{y=1/x} dx$$

$$= \int_1^\infty \left(\ln(x+1/x) - \ln x\right) dx = \int_1^\infty \ln\left(1+\frac{1}{x^2}\right) dx$$

Since, $0 < \ln(1+u) < u$ if $u > 0$, we have $0 \leq \iint_D \frac{dA}{x+y} < \int_1^\infty \frac{1}{x^2}\, dx =$

Therefore, the given integral converges a value between 0 and 1.

Ex 7 Evaluate $\iint_D \frac{1}{(x+y)^2}\, dA$ where $0 \le x \le 1$, $0 \le y \le x^2$



$$\iint_D \frac{1}{(x+y)^2}\, dA = \lim_{c\to 0^+} \int_c^1 \int_0^{x^2} \frac{dy}{(x+y)^2}\, dx$$

$$= \lim_{c\to 0^+} \int_c^1 \left(-\frac{1}{x+y}\right)\Bigg|_{y=0}^{y=x^2}$$

$$= \lim_{c\to 0^+} \int_c^1 \left(\frac{1}{x} - \frac{1}{x^2+x}\right) dx$$

$$= \int_0^1 \frac{1}{1+x}\, dx = \ln(1+x)\Big|_0^1 = \ln 2$$

NOTE 7 Nonpositive integrals could have been handled similarly. Improper integrals with integrand $f(x,y)$ that takes positive and negative values are out of scope of the course.

Definition A set D in the plane is said to be connected if any two points in it can be joined by a continuous parametric curve $x=x(t)$, $y=y(t)$ $(0 \le t \le 1)$ lying in D.

THM MEAN VALUE THEOREM FOR DOUBLE INTEGRALS

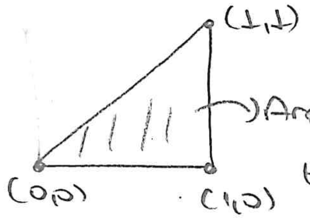
If the function $f(x,y)$ is continuous on a closed, bounded connected set D in the xy-plane, then there exists a point (x_0,y_0) in D such that

$$\iint_D f(x,y)\, dA = f(x_0,y_0)\cdot(\text{area of } D)$$

The average value or mean value of an integrable function $f(x,y)$ over the set D is the number

$$\bar{f} = \frac{1}{\text{area of } D} \iint_D f(x,y)\, dA.$$

Ex. Find average value of the integral $\iint_T (x^2+y^2)\,dA$ where T is the triagle with vertices $(0,0),(1,0),(1,1)$



Area of triagle is $=1/2$

$$\text{Average} = \frac{1}{1/2}\iint_T (x^2+y^2)\,dA$$

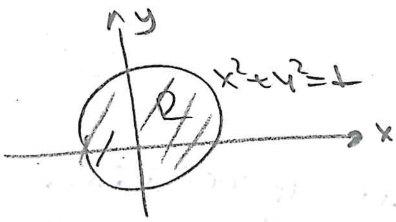
$$=2\int_0^1\int_0^x (x^2+y^2)\,dy\,dx$$

$$=2\int_0^1 \left(x^2y+\frac{y^3}{3}\right)\Big|_0^x = \frac{8}{3}\int_0^1 x^3\,dx = \frac{2}{3}$$

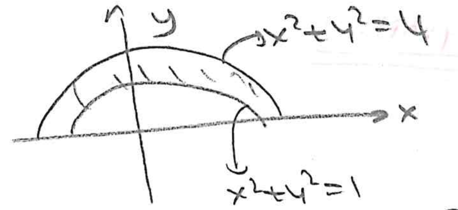
DOUBLE INTEGRALS IN POLAR COORDINATES

→ Suppose that we want to evaluate a double integral

$\iint_R f(x,y)\,dA$ where R is one of the regions as



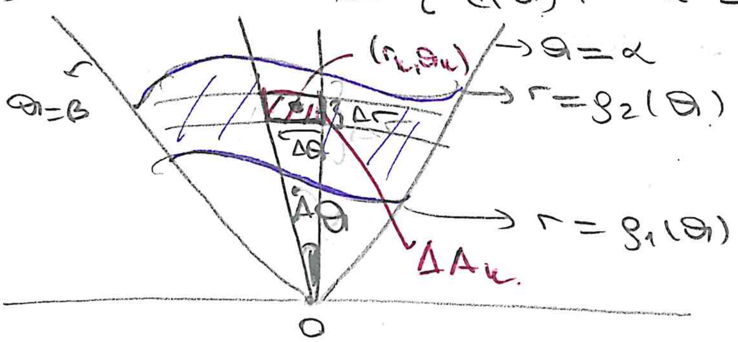
OR



$R=\{(r,\theta)\mid 0\le r\le 1,\ 0\le\theta\le 2\pi\}$ $\quad$ $R=\{(r,\theta)\mid 1\le r\le 2,\ 0\le\theta\le 2\pi\}$

In such cases, it is rather complicated to define R by using rectangular coordinates but R is easily described by using polar coordinates.

→ Consider $R=\{(r,\theta)\mid \alpha\le\theta\le\beta,\ g_1(\theta)\le r\le g_2(\theta)\}$



$$S_n=\sum_{k=1}^{n} f(r_k,\theta_k)\,\Delta A_k \xrightarrow{f \text{ is continuous}} \lim_{n\to\infty} S_n=\iint_R f(r,\theta)\,d\theta.$$

$\Delta A_k = \left(\begin{array}{c}\text{area of}\\ \text{large sector}\end{array}\right)-\left(\begin{array}{c}\text{area of}\\ \text{small sector}\end{array}\right)$

$$S_n = \sum_{k=1}^{n} f(r_k,\theta_k)\,\Delta A_k = \sum_{k=1}^{n} f(r_k,\theta_k)\, r_k\, \Delta r\, \Delta\theta$$

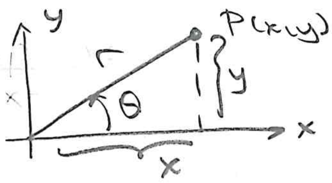
ΔA_k = (area of large sector) − (area of small sector)

$$= \frac{\Delta\theta}{2}\left[\left(r_k+\frac{\Delta r}{2}\right)^2 - \left(r_k-\frac{\Delta r}{2}\right)^2\right]$$

$$= \frac{\Delta\theta}{2}\left(2r_k\Delta r\right) = r_k\,\Delta r\,\Delta\theta$$

$$\lim_{n\to\infty} S_n = \iint_R f(r,\theta)\, r\, dr\, d\theta = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r,\theta)\, r\, dr\, d\theta$$

Recall



$r^2 = x^2 + y^2$

$x = r\cos\theta \qquad y = r\sin\theta$

Change to Polar coordinates in a double integral

If f is continuous on a polar rectangle R given by

$0 \le a \le r \le b \qquad \alpha \le \theta \le \beta$ where $0 \le \beta - \alpha \le 2\pi$, then

$$\iint_R f(x,y)\, dA = \int_{\alpha}^{\beta}\int_a^b f(r\cos\theta, r\sin\theta)\, \overbrace{r\, dr\, d\theta}^{dA}$$

Comes from Jacobian of transformation.

$dA = r\, dr\, d\theta$

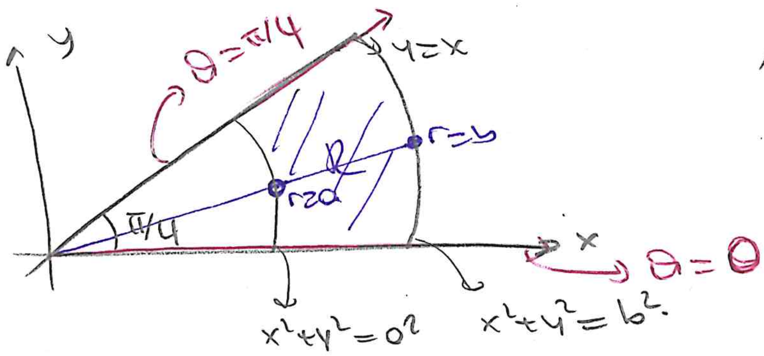
$$\underbrace{\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta}\end{vmatrix}}_{\text{Jacobian of the transformation}} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r(\cos^2\theta + \sin^2\theta) = r$$

$$dA = \left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| dr\, d\theta = |r|\, dr\, d\theta = r\, dr\, d\theta$$

Ex 1. $I = \iint_R \frac{y^2}{x^2}\, dA = ?$ R is part of the $0 < a^2 \le x^2+y^2 \le b^2$ lying in the first quadrant and below the line $y=x$

Since $y=x$

$\cancel{r}\sin\theta = \cancel{r}\cos\theta \quad \theta = \pi/4$



$I = \int_0^{\pi/4} \int_{r=a}^{r=b} \underbrace{r}_{\text{from Jacobian}} \tan^2\theta \, dr\, d\theta$

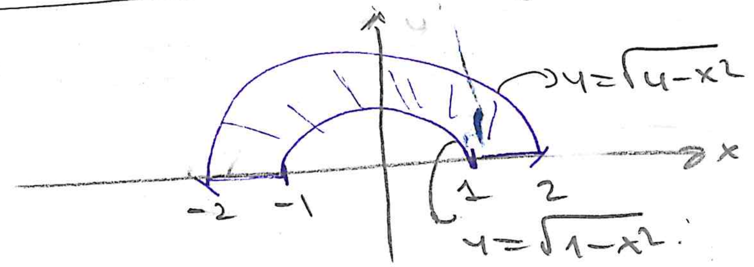
$= \int_0^{\pi/4} \tan^2\theta \left(\frac{r^2}{2}\right)\Big|_a^b d\theta$

$f(x,y) = \frac{y^2}{x^2} = \frac{r^2\sin^2\theta}{r^2\cos^2\theta} = \tan^2\theta$

$= \frac{b^2-a^2}{2} \int_0^{\pi/4} \underbrace{\tan^2\theta}_{\sec^2\theta - 1} d\theta$

$= \frac{b^2-a^2}{2} (1-\pi/4)$

Ex 2. $I = \iint_R (3x+4y^2)\, dA = ?$ where R is the region in the upper half plane bounded by circles $x^2+y^2=1$ and $x^2+y^2=4$



$R = \{(r,\theta) \mid 1 \le r \le 2,\ 0 \le \theta \le \pi\}$ $x = r\cos\theta$ $y = r\sin\theta$

$\iint_R (3x+4y^2)\, dA = \int_0^{\pi} \int_1^2 (3r\cos\theta + 4r^2\sin^2\theta)\, r \, dr\, d\theta$

$= \int_0^{\pi} \left[r^3\cos\theta + r^4\sin^2\theta\right]_1^2 d\theta = \int_0^{\pi} (7\cos\theta + 15\sin^2\theta)\, d\theta$

$= \int_0^{\pi} \left\{7\cos\theta + \frac{15}{2}(1-\cos 2\theta)\right\} d\theta = \left(7\sin\theta + \frac{15}{2}\theta - \frac{15}{2}\frac{\sin 2\theta}{2}\right)$

$= \frac{15\pi}{2}$

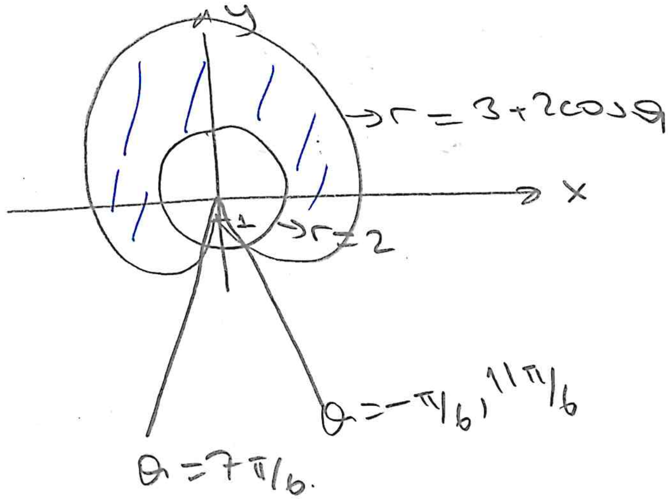
Area in Polar Coordinates

The area of closed and bounded region R in the polar coordinate plane is

$$A = \iint_R r\,dr\,d\theta$$

Ex Find the area of the region lies inside $r = 3 + 2\sin\theta$ and outside $r=2$

$$3 + 2\sin\theta = 2 \Rightarrow \sin\theta = -1/2$$
$$\theta = 7\pi/6,\ 11\pi/6$$



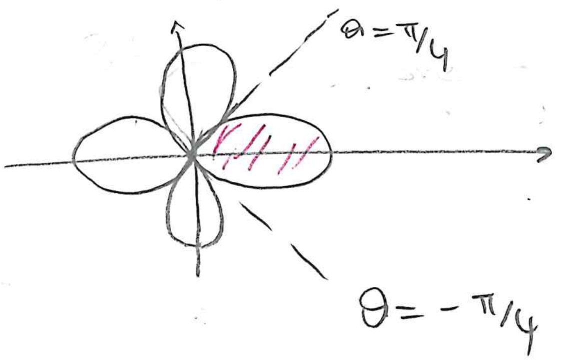
To define the shaded region we choose $-\pi/6$ instead of $11\pi/6$

Here, the region is defined by $-\pi/6 \le \theta \le 7\pi/6$, $2 \le r \le 3+2\sin\theta$

The area $A(D)$ is

$$A(D) = \iint_D dA = \int_{-\pi/6}^{7\pi/6} \int_2^{3+2\sin\theta} r\,dr\,d\theta = \frac{11\sqrt{3}}{2} + \frac{14\pi}{3}.$$

Ex Use a double integral to find the area enclosed by one-loop of the four-leaved rose $r = \cos 2\theta$



$$D = \{(r,\theta): -\pi/4 \le \theta \le \pi/4,\ 0 \le r \le \cos 2\theta\}$$

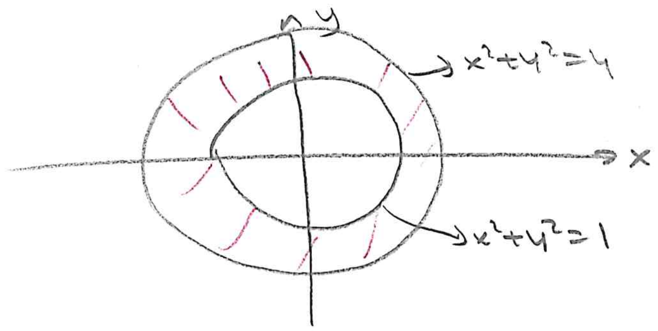
$$A(D) = \iint_D dA = \int_{-\pi/4}^{\pi/4} \int_0^{\cos 2\theta} r\,dr\,d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \frac{r^2}{2}\Big|_0^{\cos 2\theta} = \frac{1}{2}\int_{-\pi/4}^{\pi/4} \cos^2 2\theta\,d\theta$$

$$= \frac{1}{4}\int_{-\pi/4}^{\pi/4} (1 + \cos 4\theta)\,d\theta$$

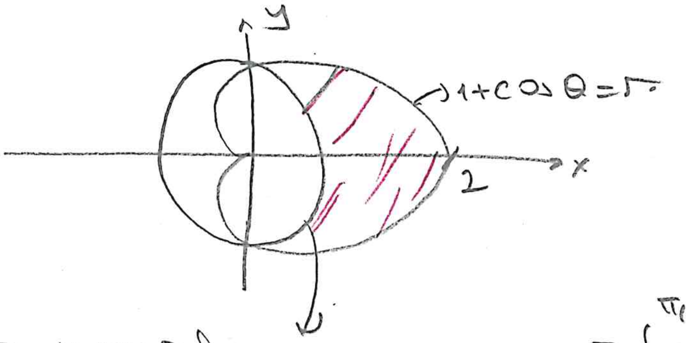
$$= \frac{1}{4}\left\{\theta + \frac{1}{4}\sin 4\theta\right\}\Big|_{-\pi/4}^{\pi/4} = \pi/8.$$

Ex Obtain the area of the region between $x^2+y^2=1$ and $x^2+y^2=4$.



$$\int_0^{2\pi}\int_{r=1}^{r=2} r\,dr\,d\theta = \int_0^{2\pi} \frac{r^2}{2}\Big|_1^2 d\theta$$
$$= \frac{3}{2}\cdot 2\pi = 3\pi$$

Ex Obtain the area of the region inside the cardioid $r=1+\cos\theta$ and outside $r=1$ circle



$r=1+\cos\theta$ & $r=1$, $r=1$

$\cos\theta=0$

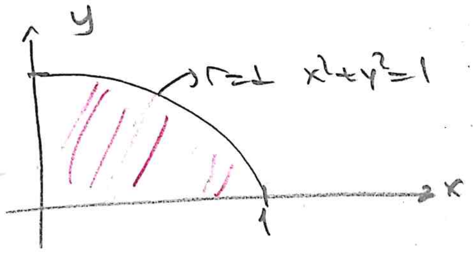
$\theta=\pi/2, -\pi/2$.

$$\int_{-\pi/2}^{\pi/2}\int_1^{1+\cos\theta} r\,dr\,d\theta = \int_{-\pi/2}^{\pi/2} \frac{r^2}{2}\Big|_1^{1+\cos\theta} d\theta$$
$$= \int_{-\pi/2}^{\pi/2} [(1+\cos\theta)^2-1]\,d\theta$$
$$= \int_{-\pi/2}^{\pi/2} (2\cos\theta+\underbrace{\cos^2\theta}_{\frac{\cos 2\theta+1}{2}})\,d\theta$$
$$= \int_{-\pi/2}^{\pi/2}\left(2\cos\theta+\frac{\cos 2\theta+1}{2}\right)d\theta$$
$$= \left(2\sin\theta+\frac{\theta}{2}+\frac{\sin 2\theta}{4}\right)\Big|_{-\pi/2}^{\pi/2} = 2+\pi/4$$

Ex $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx = ?$

Integration w.r.t y gives $\displaystyle\int_0^1 \left(x^2\sqrt{1-x^2}+\frac{(1-x^2)^{3/2}}{3}\right)dx$

Change to polar coordinates.

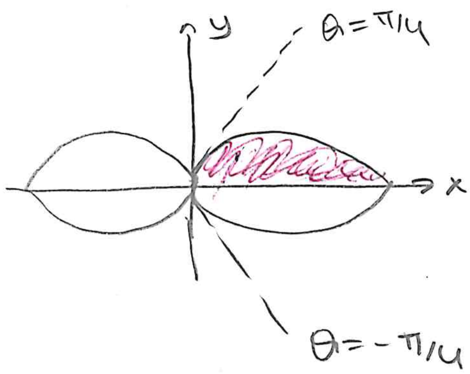


$y=0$ $y=\sqrt{1-x^2}$

$x=0$ $x=1$.

$$\int_0^1\int_0^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx = \int_0^{\pi/2}\int_0^1 r^2\, r\,dr\,d\theta$$
$$= \int_0^{\pi/2} \frac{r^4}{4}\Big|_0^1 d\theta = \pi/8.$$

Ex: Find the area enclosed by the lemniscate $r^2 = 4\cos 2\theta$



$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta$$

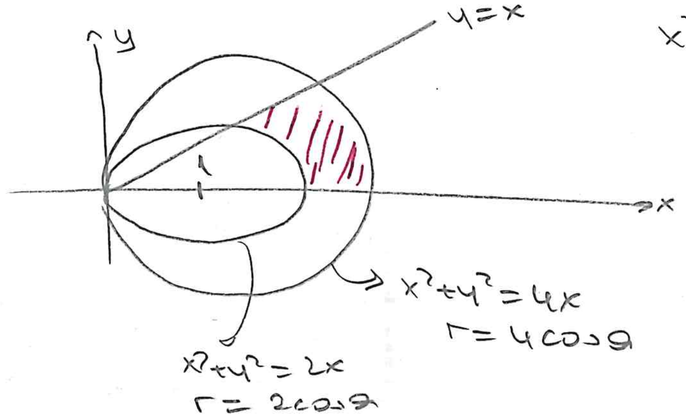
$$= 4\int_0^{\pi/4} \frac{r^2}{2}\Big|_0^{\sqrt{4\cos 2\theta}} d\theta$$

$$= 4\int_0^{\pi/4} 2\cos 2\theta\, d\theta$$

$$= 4\sin 2\theta \Big]_0^{\pi/4} = 4$$

Ex Find the area of the region bounded by $x^2+y^2 = 2x$

$x^2+y^2=4x$, $y=x$ and $y=0$



$x = r\cos\theta \qquad y = r\sin\theta$

$x^2+y^2 = 2x \Rightarrow r^2 = 2r\cos\theta$

$r = 2\cos\theta$

$x^2+y^2 = 4x \Rightarrow r^2 = 4r\cos\theta$

$r = 4\cos\theta$

$x = y$

$\cancel{r}\cos\theta = \cancel{r}\sin\theta$

$\theta = \pi/4$

$$\int_0^{\pi/4}\int_{2\cos\theta}^{4\cos\theta} r\,dr\,d\theta = \int_0^{\pi/4} \frac{r^2}{2}\Big|_{2\cos\theta}^{4\cos\theta} d\theta = \int_0^{\pi/4} \frac{16\cos^2\theta - 4\cos^2\theta}{2}\,d\theta$$

$$= 6\int_0^{\pi/4} \underbrace{\cos^2\theta}_{\frac{1+\cos 2\theta}{2}}\,d\theta = 6\left(\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right)\Big|_0^{\pi/4} = 3\left(\pi/4 + \frac{1}{2}\right)$$

Ex Write the following integrals using polar coordinates

① $I = \int_{-1}^{1}\int_0^{\sqrt{1-x^2}} dy\,dx = ?$

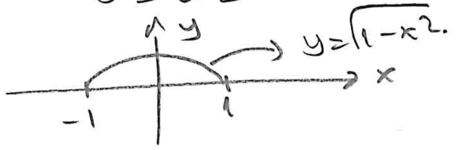
$$I = \int_0^{\pi}\int_0^{1} r\,dr\,d\theta = \int_0^{\pi} \frac{r^2}{2}\Big|_0^1 d\theta = \pi/2.$$

$y=0 \qquad y = \sqrt{1-x^2}$

$x=-1 \qquad x=1$

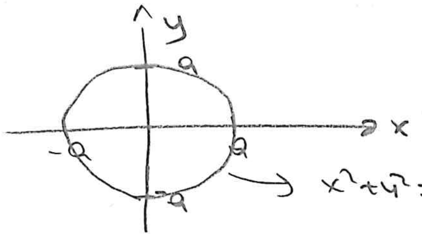
$0 \le r \le 1$

$0 \le \theta \le \pi$



(2) $I = \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\,dx = ?$

$$I = \int_0^a \int_0^{2\pi} r\,d\theta\,dr = \int_0^a 2\pi r\,dr = \pi a^2.$$



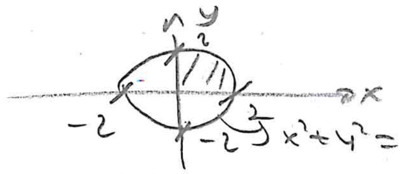
$\to x^2+y^2=a^2 \qquad y=-\sqrt{a^2-x^2} \qquad y=\sqrt{a^2-x^2}$

$-a \le x \le a$

(3) $$\int_0^2 \int_0^{\sqrt{4-y^2}} (x^2+y^2)\,dx\,dy = \int_0^{\pi/2} \int_0^2 r^2 r\,dr\,d\theta = \frac{2^4}{4}\cdot \pi/2 = 2\pi$$

$x=0 \qquad x=\sqrt{4-y^2} \Rightarrow x^2+y^2=4$

$y=0 \qquad y=2.$

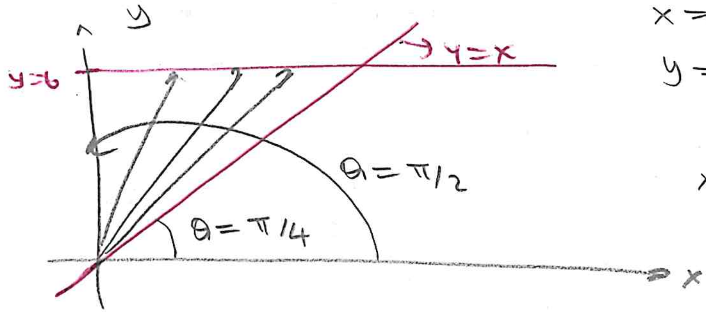


$\to x^2+y^2=4$

$0 \le \theta \le \pi/2$

$0 \le r \le 2.$

(4) $$\int_0^6 \int_0^4 x\,dx\,dy = \int_{\pi/4}^{\pi/2} \int_0^{6/\sin\theta} r\; r\cos\theta\,dr\,d\theta$$



$x=0 \qquad x=4$

$y=0 \qquad y=6 \to r\sin\theta=6$

$r=6/\sin\theta$

$x=4 \Rightarrow r\cos\theta = r\sin\theta$

$\theta=\pi/4$

$$\int_{\pi/4}^{\pi/2} \frac{r^3}{3}\cos\theta \Big|_0^{6/\sin\theta} d\theta = \int_{\pi/4}^{\pi/2} \frac{216}{3}\frac{\cos\theta}{\sin^3\theta}\,d\theta = \int_{\pi/4}^{\pi/2} \frac{216}{3} \underbrace{\frac{\cos\theta}{\sin\theta}}_{\cot\theta} \cdot \underbrace{\frac{1}{\sin^2\theta}}_{\mathrm{cosec}^2\theta}\,d\theta$$

$$= -\frac{216}{3}\,\frac{\mathrm{cosec}^2\theta}{2}\Big|_{\pi/4}^{\pi/2} = -\frac{216}{6}\left(\frac{1}{\sin\theta}\right)^2 \Big|_{\pi/4}^{\pi/2} = 36$$

$\mathrm{cosec}\,\theta = u \qquad (\mathrm{cosec}\,\theta)' = -\cot\theta\,\mathrm{cosec}\,\theta$

$-\cot\theta\,\mathrm{cosec}\,\theta\,d\theta = du \qquad -du$

$$\int \underbrace{\cot\theta\cdot\mathrm{cosec}\,\theta\;d\theta}_{-du}\;\underbrace{\mathrm{cosec}\,\theta}_{u} = -\frac{u^2}{2}+C.$$